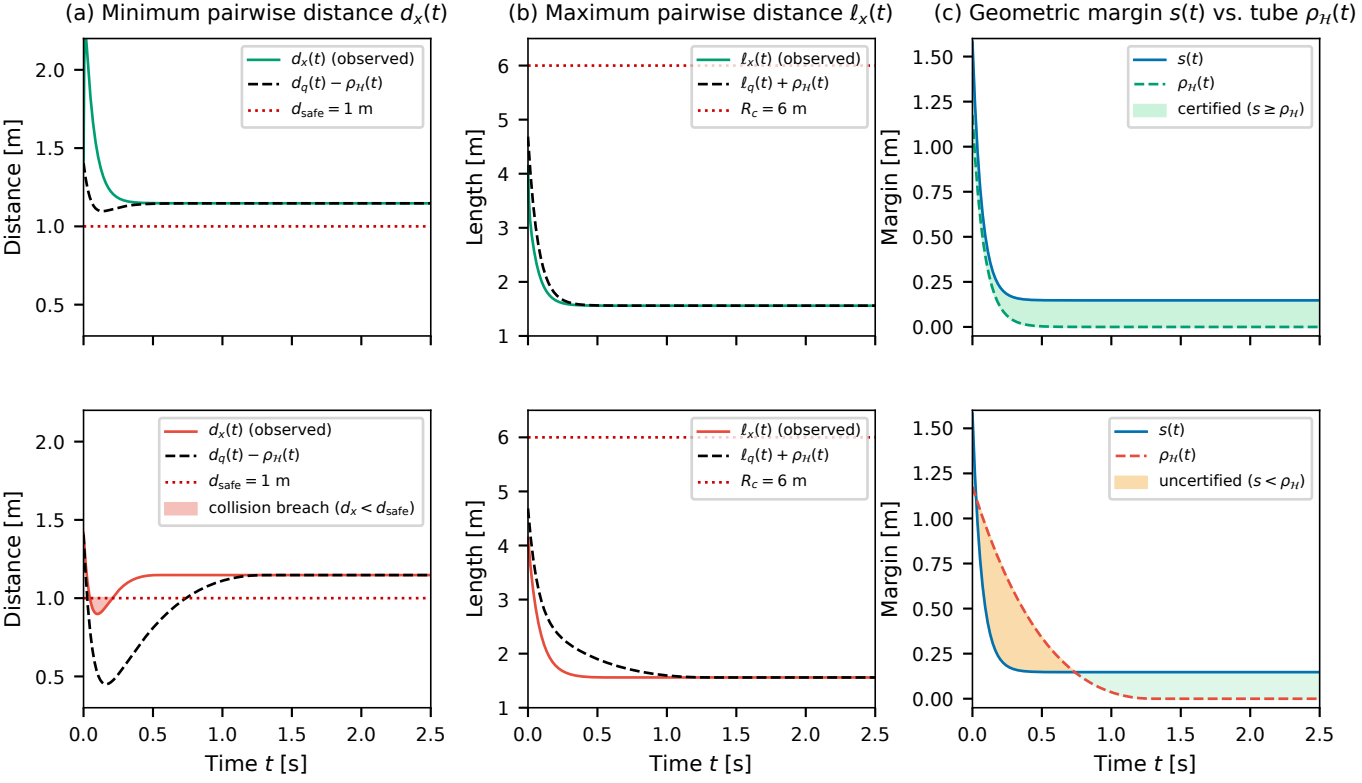

(a) Minimum pairwise distance $d_x(t)$
(b) Maximum pairwise distance $\ell_x(t)$
(c) Geometric margin $s(t)$ vs. tube $\rho_{\mathcal{H}}(t)$
Distance [m]
Length [m]
Margin [m]
Time $t$ [s]
$d_x(t)$ (observed)
$d_q(t) - \rho_{\mathcal{H}}(t)$
$d_{\text{safe}} = 1$ m
$\ell_x(t)$ (observed)
$\ell_q(t) + \rho_{\mathcal{H}}(t)$
$R_c = 6$ m
$s(t)$
$\rho_{\mathcal{H}}(t)$
certified ($s \geq \rho_{\mathcal{H}}$)
collision breach ($d_x < d_{\text{safe}}$)
uncertified ($s < \rho_{\mathcal{H}}$)